

JOSHUA S. SPEAGLE (沈佳士)

Statistical Sciences | Astronomy & Astrophysics

University of Toronto

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RESEARCH INTERESTS

My research lies at the intersection of **astrostatistics**, **data science**, and **artificial intelligence**. I develop and apply a variety of methods to **large datasets** to answer complex astrophysical questions, with a focus on the formation, behaviour, and evolution of **stars** and **galaxies**.

POSITIONS

Assistant Professor of Astrostatistics: University of Toronto	2022-Present
Dunlap Postdoctoral Fellow: University of Toronto	2020-2022
Banting Postdoctoral Fellow: University of Toronto	2020-2022
Supervisor: Gwen Eadie	
Project Academic Support Staff: Kavli IPMU, University of Tokyo	2015-2016
Supervisors: Naoki Yoshida, Alexie Leauthaud, & Kevin Bundy	

EDUCATION

Harvard University: PhD in Astronomy	2016-2020
Advisers: Doug Finkbeiner, Charlie Conroy, & Daniel Eisenstein (with Alyssa Goodman)	
Harvard University: MA in Astronomy	2016-2020
Advisers: Daniel Eisenstein & Alexie Leauthaud	
Harvard University: BA in Astrophysics and Physics	2011-2015

SELECTED AWARDS & HONORS

Connaught New Researcher Award (University of Toronto)	2024
John Charles Polanyi Prize in Physics (Council of Ontario Universities)	2022
Best Astrostatistics Student Paper Award (ASA/AIG)	2020
Eric R. Keto Prize for Best Thesis in Theoretical Astrophysics (Harvard)	2020

TEACHING

I have a strong interest in education and pedagogy, with a focus on skills such as programming, applied statistics, and data science. See my [teaching statement](#) for additional details.

EQUITY, DIVERSITY, & INCLUSION

I am committed to improving equity, diversity, and inclusion (EDI) in the classroom, in my work, and in the wider academic community. See my [EDI statement](#) for additional details.

SUPERVISION & MENTORSHIP

I am currently (co-)supervising/mentoring a total of **16 individuals**. This includes:

- 6 postdoctoral researchers
- 7 doctoral students
- 3 undergraduate students

See my [List of Mentees](#) for a full record of the **65 individuals** I have (co-)supervised/mentored.

SELECTED PROFESSIONAL ACTIVITIES & SERVICE

American Astronomical Society (AAS)

Working Group on Astroinformatics & Astrostatistics Steering Committee 2020-Present

American Statistical Association (ASA) Astrostatistics Interest Group (AIG)

Program Chair (Interim) 2023-2024

Web Director 2020-2023

Canadian Astronomical Society (CASCA)

Postdoc Committee 2020-2023

Statistical Society of Canada (SSC)

Data Science and Analytics Section: President Elect 2024-2025

LSST Informatics & Statistics Science Collaboration

Publication Committee: Chair 2024

University of Toronto (UofT) Astronomy

Undergraduate Curriculum Committee 2023-2024

Summer Undergraduate Research Program (SURP) Committee: Chair 2022-2024

Summer Undergraduate Research Program (SURP) Committee 2021-2022

Training & Mentoring Committee: Co-Chair 2021-2023

Graduate Admissions Committee 2021

Center for Astrophysics | Harvard & Smithsonian (CfA)

CfA Machine Learning Journal Club: Founder and Organizer 2017-2020

Workshops & Conferences

Annual UofT STATSTRO Workshop: Co-Organizer 2021-Present

Globular Clusters & Their Tidal Tails: SOC Co-Chair, LOC Co-Chair 2024

CASCA Annual General Meeting: SOC Co-Chair, LOC Member 2024

Joint Statistical Meetings (JSM) 2021 2021

Topic-Contributed Panel: Understanding a Data-Rich Universe with Data-Driven Approaches

Manuscript Referee

Annals of Applied Statistics 2022-Present

Statistical Science 2022-Present

RAS Techniques and Instruments (RASTI) 2022-Present

Bayesian Analysis 2021-Present

Journal of Open Source Software (JOSS) 2020-Present

Astronomy & Astrophysics (A&A) 2017-Present

Monthly Notices of the Royal Astronomical Society (MNRAS) 2016-Present

American Astronomical Society (AAS) Journals (AJ, ApJ, ApJL, ApJS) 2014-Present

SELECTED PRESENTATIONS & APPEARANCES

I have given numerous presentations related to work being done by me and my collaborators to the scientific community and the public along with providing lectures and tutorials and participating in panel discussions. Recent appearances are provided below. See my [List of Talks](#) for a full record.

Invited Talks

AstroAI Summer Workshop	Jun 2024
AI SpAI with My (Little?) AI: Applications of Statistical Learning in Modern Astrophysics	
NSF IAIFI Summer Workshop	Aug 2023
Uncertainty and Interpretability in Machine Learning Models	
MaxEnt 2023 Workshop	Jul 2023
Frontiers of Nested Sampling Meeting-within-a-Meeting	
Musings on Nested Sampling from Interactions with Everyday Users	
Statistical Challenges in Modern Astronomy VIII	Jun 2023
Extending Bayesian Modeling Approaches to ML Applications	
IAU General Assembly 2022	Aug 2022
Machine Learning in Astronomy: Possibilities and Pitfall	
Incorporating Errors into Machine Learning Methods	

Colloquia & Seminars

Tsinghua University	Dec 2024
Colloquium	
“Trust Me, I am an Astrophysics Foundation Model”	
NRAO	Jun 2024
NRAO/GBO REU Seminar	
From “Chi by Eye” to MCMC: An Introduction to Astrostatistics and Estimating Uncertainties in Astronomy	
Ohio State University	Mar 2024
CCAPP Seminar	
I Spy with my (Little?) AI: Applications of Statistical Learning in Modern Astrophysics	
Université de Montréal	Oct 2023
Astrophysics Seminar	
Astrostatistical Thinking in Machine Learning Applications	
Carnegie Mellon University	Sep 2023
STAMPS Seminar	
Nested Sampling: Past, Present, and Future	
University of Pittsburgh	Sep 2023
Astronomy Seminar	
From Panchromatic Images to Stellar Ages and Beyond	
Center for Astrophysics Harvard & Smithsonian	Aug 2023
AstroAI Seminar	

Extending Statistical Thinking to ML Applications	
McMaster University	Apr 2023
Astronomy Lunch Seminar	
From Panchromatic Images to Stellar Ages and Beyond	
University of Waterloo	Feb 2023
Astroseminar	
From Panchromatic Images to Stellar Ages and Beyond	
NRC Herzberg Astronomy and Astrophysics Research Centre	Nov 2022
Colloquium	
Galaxy Evolution through the Eyes of the Milky Way	
Queen's University	Nov 2022
Astronomy Seminar	
Galaxy Evolution through the Eyes of the Milky Way	
Institute for Advanced Study	Aug 2022
Astrophysics Seminar	
From Panchromatic Images to Stellar Ages and Beyond	
Saint Mary's University	Aug 2022
Department of Astronomy and Physics Colloquium	
More Data + More Parameters = More Fun with Galaxy Evolution	

Contributed Talks

Banff International Research Station	Nov 2023
Astrostatistics in Canada and Beyond	
SBI++: Extending Simulation-Based Inference to Censored and Out-of-Distribution Data	
University of Michigan	Aug 2023
2023 Clusters & Streams Workshop	
Rapid Stellar Parameter Inference with Probabilistic Machine Learning	
JSM 2023	Aug 2023
True North Strong and... Amazing at Astrostatistics! (Invited Paper Session)	
Connecting Galaxy Evolution between Observation, Simulation, and Theory	
Surveys to Discovery (S2D) Conference	May 2023
Photometric Biases in Modern Surveys	
2023 Spring Research Conference (SRC)	May 2023
From Light to Mass and Back Again: Exploring Galaxy Evolution with “New” Methods and “Old” Outcomes	
McMaster University	Aug 2022
2022 Clusters Workshop	
Panchromatic Modelling of Co-Eval Stellar Populations	
JSM 2022	Aug 2022
Advances in Astrostatistics in the Great White North (Invited Paper Session)	
Mapping the Milky Way in 5-D with Big Data	

Lectures & Tutorials

Summer School in Statistics for Astronomers Nested Sampling: Past, Present, and Future	Jun 2024
2023 ESO Conference Two in a million – The interplay between binaries and star clusters Machine Learning for Star Cluster Science	Sep 2023
University of Michigan 2023 Clusters & Streams Workshop Dimensionality Reduction: Applications to Star Cluster Science	Aug 2023
NSF IAIFI Summer School Uncertainty and Interpretability in Machine Learning Models (with Alex Gagliano)	Aug 2023
Statistical Challenges in Modern Astronomy VIII A Tutorial of (Dynamic) Nested Sampling	Jun 2023

Panels

NeurIPS 2023 Workshop on Machine Learning and the Physical Sciences Panel on inductive biases and interpretability	Dec 2023
NSF IAIFI Summer School Career Panel	Aug 2023
Dunlap Institute for Astronomy & Astrophysics 15 th Anniversary Homecoming Celebration From Dunlap to Diverse Careers: A Panel Discussion on Opportunities and Challenges for Astrophysics Graduates (Moderator)	Jun 2023

Posters

2023 ESO Conference Two in a million – The interplay between binaries and star clusters A machine-learning led search for extra-tidal stars of globular clusters	Sep 2023
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Public Talks & Events

Astronomy on Tap (Toronto) Host	Jul 2024
Christie Gardens Apartments & Care (Toronto) The Role of AI in Modern Astrophysics	Nov 2023
Astronomy on Tap (Toronto) Mapping the Milky Way from the Inside Out	Nov 2022

PUBLICATIONS

I am an author on **100+ papers** that have **11000+** citations ([h-index=40](#)). This includes:
10+ papers as (co-)lead author with 3000+ citations ([h-index=11](#))

35+ papers with significant contributions with 3000+ citations ([h-index=18](#))

15+ papers led by students with 100+ citations ([h-index=5](#))

Most of my papers can be found online on [ADS](#), [arxiv](#), and [Google Scholar](#). My ORCID is [0000-0003-2573-9832](#). See my full [Publications List](#) for additional details. Citation numbers from ADS.

Top 5 Most Cited Publications as (Co-)Lead Author

- 1497 cites: **Speagle, J. S.**, 2020, Monthly Notices of the Royal Astronomical Society, Vol. 493, Iss. 3, p. 3132-3158
dynesty: A Dynamic Nested Sampling Package for Estimating Bayesian Posteriors and Evidences
arxiv: [1904.02180](#)
- 1344 cites: **Speagle, J. S.**; Steinhardt, C. L.; Capak, P. L.; & Silverman, J. D., 2014, The Astrophysical Journal Supplement Series, Vol. 214, Iss. 2, id. 15
A Highly Consistent Framework for the Evolution of the Star-Forming ‘Main Sequence’ from $z \sim 0-6$
arxiv: [1405.2041](#)
- 217 cites: **Zucker, C. & Speagle, J. S.**; Schlafly, E. F.; Green, G. M., Finkbeiner, D. P.; Goodman, A. A.; & Alves, J., 2019, The Astrophysical Journal, Vol. 879, Iss. 2, id. 125
A Large Catalog of Accurate Distances to Local Molecular Clouds: The Gaia DR2 Edition
arxiv: [1902.01425](#)
- 44 cites: **Speagle, J. S.**, arxiv e-print
A Conceptual Introduction to Markov Chain Monte Carlo Methods
arxiv: [1909.12313](#)
- 30 cites: **Speagle, J. S.** et al. [11 additional co-authors], 2019, Monthly Notices of the Royal Astronomical Society, Vol. 490, Iss. 4, p. 5658-5677
Galaxy-Galaxy Lensing in HSC: Validation Tests and the Impact of Heterogeneous Spectroscopic Training Sets
arxiv: [1906.05876](#)

Top 5 Most Cited Publications with Significant Contributions

- 950 cites: Green, G. M.; Schlafly, E. F.; Zucker, C.; **Speagle, J. S.**; & Finkbeiner, D. P., 2019, The Astrophysical Journal, Vol. 887, Iss. 1, id. 93
A 3D Dust Map Based on Gaia, Pan-STARRS 1 and 2MASS
arxiv: [1905.02734](#)
- 487 cites: Johnson, B. D.; Leja, J.; Conroy, C.; & **Speagle, J. S.**, 2021, The Astrophysical Journal Supplement Series, Vol. 254, Iss. 2, id. 22
Stellar Population Inference with Prospector
arxiv: [2012.01426](#)
- 397 cites: Leja, J.; Carnall, A. C.; Johnson, B. D.; Conroy, C.; & **Speagle, J. S.**, 2019, The Astrophysical Journal, Vol. 876, Iss. 1, id. 3
How to Measure Galaxy Star Formation Histories II: Nonparametric Models
arxiv: [1811.03637](#)

- 275 cites: Tanaka, M.; Coupon, J.; Hsieh, B.-C.; Mineo, S.; Nishizawa, A. J.; **Speagle, J.**; Furusawa, H.; Miyazaki, S.; & Murayama, H., 2018, Publications of the Astronomical Society of Japan, Vol. 70, Iss. SP1, id. S9
Photometric Redshifts for the Hyper Suprime-Cam Subaru Strategic Program Data Release 1
arxiv: [1704.05988](https://arxiv.org/abs/1704.05988)
- 182 cites: Zucker, C.; **Speagle, J. S.**; Schlafly, E. F.; Green, G. M.; Finkbeiner, D. P.; Goodman, A.; & Alves, J., 2020, Astronomy & Astrophysics, Vol. 633, id. A51
A compendium of distances to molecular clouds in the Star Formation Handbook
arxiv: [2001.00591](https://arxiv.org/abs/2001.00591)

Top 5 Most Cited Publications as a Contributing Author

- 932 cites: Aihara, H. et al. [142 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, Iss. SP1, id. S4
The Hyper Suprime-Cam SSP Survey: Overview and Survey Design
arxiv: [1704.05858](https://arxiv.org/abs/1704.05858)
- 566 cites: Aihara, H. et al. [108 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, Iss. SP1, id. S8
First Data Release of the Hyper Suprime-Cam Subaru Strategic Program
arxiv: [1702.08449](https://arxiv.org/abs/1702.08449)
- 537 cites: Hikage, C. et al. [35 additional co-authors including **Speagle, J. S.**], 2019, Publications of the Astronomical Society of Japan, Vol. 71, Iss. 2, id. 43
Cosmology from cosmic shear power spectra with Subaru Hyper Suprime-Cam first-year data
arxiv: [1809.09148](https://arxiv.org/abs/1809.09148) **Media:** [PASJ Excellent Paper Award \(English\)](#)
- 227 cites: Mandelbaum, R. et al. [30 additional co-authors including **Speagle, J. S.**], 2018, Publications of the Astronomical Society of Japan, Vol. 70, Iss. Sp1, id. S25
The first-year shear catalog of the Subaru Hyper Suprime-Cam SSP Survey
arxiv: [1706.06745](https://arxiv.org/abs/1706.06745)
- 203 cites: Leja, J.; Johnson, B. D.; Conroy, C.; van Dokkum, P.; **Speagle, J. S.**; Brammer, G.; Momcheva, I.; Skelton, R.; Whitaker, K. E.; Franx, M.; & Nelson, E. J., 2019, The Astrophysical Journal, Vol. 877, Iss. 2, id. 140
An Older, More Quiescent Universe from Panchromatic SED Fitting of the 3D-HST Survey
arxiv: [1812.05608](https://arxiv.org/abs/1812.05608)